Higher National Diploma in Information Technology

Second Year, First Semester Examination – 2018

HNDIT2302 IT Project Management

Instructions for Candidates:
Answer any **04 Questions**
All questions carry equal marks.
Only non-programmable calculators are allowed.

No. of questions : 05
No. of pages : 03
Time : Two (02) hours

Q1)

i. Define the term **Project Management**. (03 Marks)

ii. Name FOUR criteria by which a project can be judged as a successful project. (04 Marks)

iii. Project management can be viewed as a number of interlinked processes. Name those five processes. (05 Marks)

iv. List five (05) advantages of using formal Project Management. (05 Marks)

v. IT project managers are expected to possess certain skills. Describe **four** skills of good IT project managers. (08 Marks)

[Total 25 Marks]

Q2)

i. What is Project Human Resource Management? (03 Marks)

ii. "Good project integration management is the key to overall project success". Explain the validity of this statement. (04 Marks)

iii. There is usually not enough time or resources to implement all available projects. What are the suitable methods to select a project? (05 Marks)

v. C&N Company is considering two mutually exclusive projects, project A and project B. The cash flows for each would be as follow. All figures are given in Sri Lankan Rupees.

Year	Discount factor	Project A	Project B
0	1.000	(10,900)	(35,250)
1	0.862	6,000	18,000
2	0.743	5,000	15,000
3	0.641	3,000	15,000

The company's cost of capital is 16%.

a. Calculate the Net Present Value of each project. (06 Marks)

b. Give your recommendation and reasons as to which project should be accepted. (04 Marks)

v. List two advantages of the Discount Factor (DCF) method of project appraisal over other appraisal methods. (03 Marks)

[Total 25 Marks]

Q3)

i. Explain the term Work Breakdown Structure (WBS) and give two approaches to Develop WBS. (04 Marks)

ii. SLIATE management has decided to buy in and install new software package to replace the existing in-house system to prepare examination results. Assume that you are the project manager and have drawn up an outline project plan for the main project activities, as follows:

Activity	Predecessor	Duration (weeks)
A	-	4
B	A	4
C	A	12
D	B	4
E	D	4
F	C	4
G	C,D	6
H	E,F,G	2

a. Draw a network diagram for the project. (05 Marks)

b. On the diagram, compute Early Start (ES), Early Finish (EF), Late Start (LS) and Late Finish (LF) time estimates and total float for each activity. (08 Marks)

c. Find the critical path and project duration. (06 Marks)

iii. Write advantages of using an Activity on Node network diagram over an activity-on-arrow network diagram in project planning. (02 Marks)

[Total 25 Marks]

Q4)

Your company has decided to develop a new information system.

i. Define the term "stakeholder" in relation to an IT development project. (03 Marks)

ii. "Active user involvement is a key in satisfying user requirements". Do you agree with this statements? Justify your answer. (04 Marks)

iii. Describe the purpose of following documents in project management process.

a. Project charter

b. Project scope statement (05 Marks)

iv. Suppose that a software project was estimated to be 400 000 line of codes. Calculate the effort and development time for organic mode using basic COCOMO equation.

Hint: Effort (E) $= a * (size)^b$,

Development Time $= c * E^d$;

Where a, b, c and d equals to 2.4, 1.05, 2.5 and 0.38 (05 Marks)

v. Briefly explain the typical problems of cost estimates in IT projects. (08 Marks)

[Total 25 Marks]

Q5)

i. What is meant by **quality** in project management? (03 Marks)

ii. State the most significant characteristics of effective and ineffective project managers. Give two points per each. (04 Marks)

iii. What are the differences between quality assurance and quality control (05 Marks)

iv. Briefly describe the impact of using modern software tools and techniques in project communications. (05 Marks)

v. Briefly explain the following terms.

a. Project Risk Management

b. Project Procurement Management (08 Marks)

[Total 25 Marks]